MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, 0.086 DIAMETER, SEMIRIGID, 50 OHMS, M17/133-RG405, M17/133-00001 THROUGH -00003

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

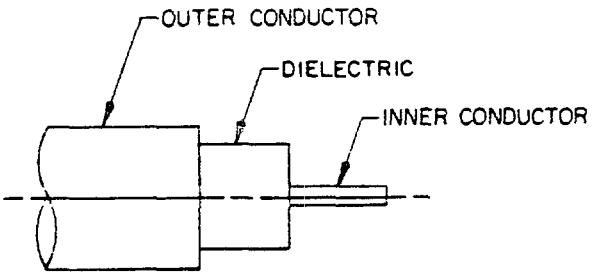


FIGURE 1. Configuration.

TABLE I. Description.

Part number	Inner conductor	Dielectric core	Outer conductor	Weight, per 1000 feet (maximum)
M17/133-RG405	Solid, silver-coated, copper clad steel .0201 inch ±.0005	Type F-1 diameter .066 inch ±.002	Seamless copper tubing, diameter .0865 inch ±.001	15.3
M17/133-00001	Same as above	Same as above	Seamless copper tubing tin-plated per MIL-T-10727, Type I, .0003 to .0005 inch thick, diameter .0865 inch +.002, -.001	15.8
M17/133-00002	Solid, silver-coated, copper wire, diameter .0201 inch ±.0005	Same as above	Seamless copper tubing diameter .0865 inch ±001	15.2
M17/133-00003	Same as above	Same as above	Seamless copper tubing tin-plated per MIL-T-10727, type I, .0003 to .0005 inch thick, diameter .0865 inch +.002, -.001	15.7

FSC 6145

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Engineering information:

Continuous working voltage: 1,500 Vrms, maximum.
Operating frequency: 20 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -40° to +125°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 2.57 ohms per 100 feet for class A wire.
6.55 ohms per 100 feet for class HS wire.
Elongation: 15 percent minimum for class 40 A wire.
1.0 percent minimum for class 40 HS wire.
Engineering notes: This cable useful in critical RF performance applications (see connector series SMA per MIL-C-39012). This cable manufactured in 20-foot lengths.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Eccentricity: 8.5 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 4 pounds minimum, 25 pounds maximum.
Aging stability: Not applicable.
Stress crack resistance: Not applicable.
Outer conductor integrity: +75° ±5°C.
Cold bend: Not applicable.
Dimensional stability: +125° ±5°C.
Core to jacket: .015 inch, maximum.
Contamination: Not applicable.
Bendability: Mandrel diameter: .25 inch, maximum.
Flammability: Not applicable.
Weight: See table I.

Electrical:
Test frequency: 500 MHz to 20 GHz.
Spark test: Not applicable.
Voltage withstanding: 5,000 Vrms.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,500 Vrms, minimum.
Characteristic impedance: 50.0 ohms ±1.5.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 32 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Phase stability: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/133-RG405	RG-405/U
M17/133-00001	---
M17/133-00002	---
M17/133-00003	---

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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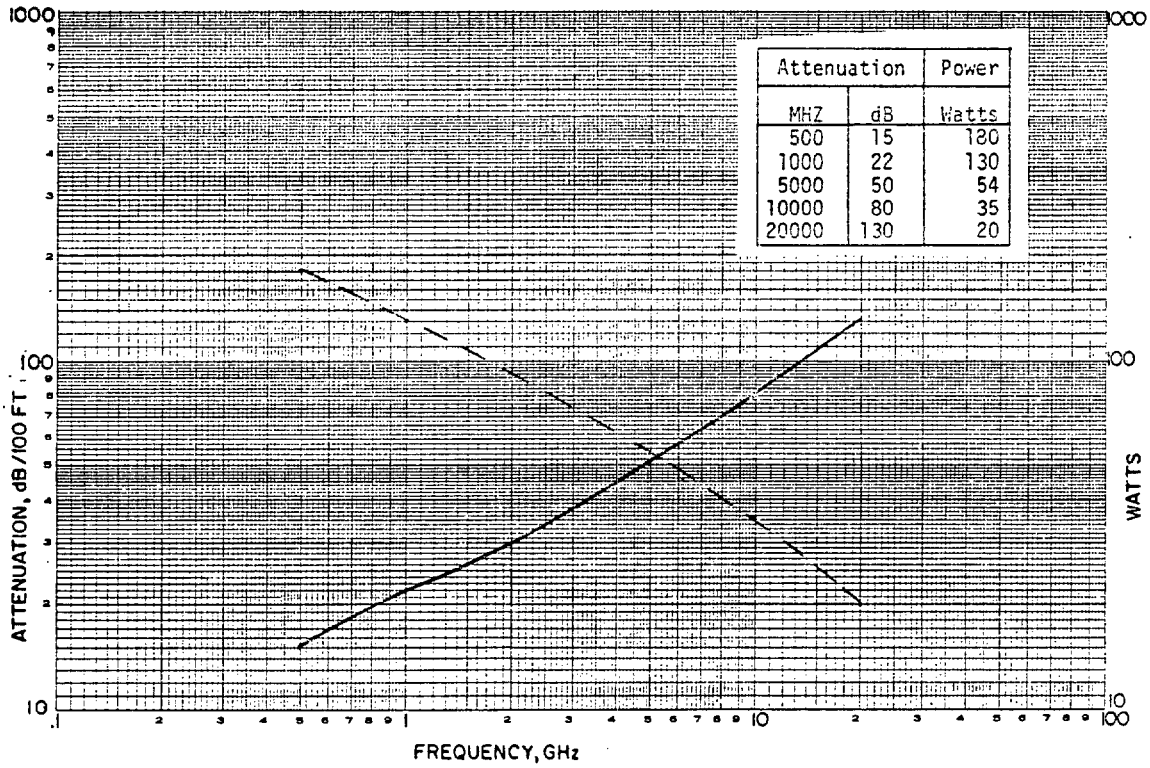


FIGURE 2. Power rating and attenuation.

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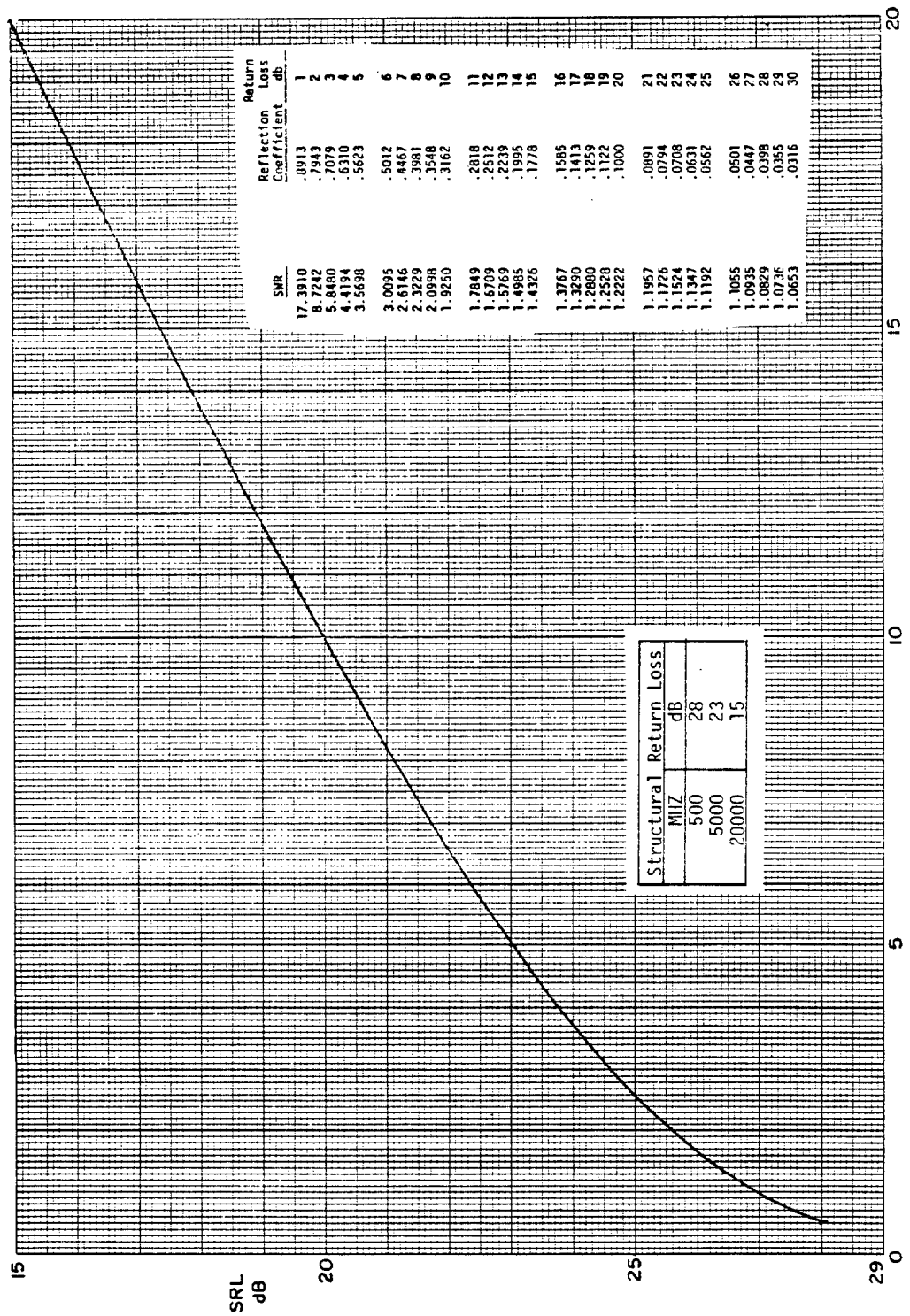


FIGURE 3. Minimum structural return loss cable.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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